Chronic Lymphocytic Leukemia

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85.1 Introduction

CLL is a rare indication for HCT since it usually follows an indolent course which does not require treatment in the majority of patients. In case of treatment indication, numerous pathway inhibitor therapy regimens are available, essentially based on Bruton's-Tyrosine-Kinase inhibitors (BTKi) or the BCL2 inhibitor venetoclax with or without CD20 antibodies. The high efficacy and the non-cross-resistance of the available chemotherapy-free treatment options offers chances for long-term survival although they cannot provide cure. Only a small minority of patients will follow an unfavorable course on pharmacological treatment and may benefit from cellular immunotherapy.

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85.2 Principles of Treatment for CLL

Since CLL is not curable with standard therapy, the diagnosis of CLL does not justify immediate start of treatment. Criteria that trigger treatment are symptomatic disease, progressive anemia or thrombocytopenia, a lymphocyte doubling time of less than 6 months, constitutional symptoms, massive splenomegaly or lymphadenopathy, and Richter transformation. Labeled first-line regimens comprise (1) covalent BTKi (Acalabrutiib, Zanubrutinib, Ibrutinib) with or without CD20 antibody loading phase; (2) Venetoclax with the CD20 antibody obinutuzumab; and (3) the combination of ibrutinib and venetoclax. Whereas (2) and (3) are administered for a fixed duration (12–15 months), (1) is given permanently until intolerability or CLL resistance. With all of these options, high response rates (>80%) and long-lasting responses (median duration of response beyond 5 years) can be achieved (Sharman et al. 2022; Al-Sawaf et al. 2023). Patients with CLL harboring a cytogenetic deletion 17p or with a *TP53* mutation detected by DNA-sequencing (combined in this manuscript as *TP53* abnormalities) as well as those with a complex karyotype may follow a less favorable course (Al-Sawaf et al. 2023; Huber et al. 2023; Furstenau et al. 2023).

If first-line therapy fails and a secondary treatment indication emerges, the general therapeutic principle is a class switch (i.e., 1 L BTKi - > 2 L

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A. Sureda et al. (eds.), *The EBMT Handbook*, https://doi.org/10.1007/978-3-031-44080-9_85

Venetoclax and vice versa; 1 L fixed duration with 2 L permanent covalent BTKi). Patients failing on both classes of pathway inhibitors have a poor outlook (Mato et al. 2020). It is unclear if the recent advent of noncovalent BTKi can improve the adverse prognosis of these "double refractory" patients (Mato et al. 2023).

85.3 Allogeneic HCT for CLL

Accordingly, double refractory patients as well as those having failed one pathway inhibitor class in the presence of *TP53* abnormalities and/or complex karyotype should be considered for alloHCT (Dreger 2021). Patients having received a BTKI/Venetoclax fixed-duration combination as first-line therapy must not be considered as double refractory if treatment failure occurs after discontinuation of the BTKi. Regardless of pathway inhibitor exposure transplant-eligible patients with a history of Richter's transformation have an indication of alloHCT.

Available evidence strongly suggests that alloHCT is currently the only therapy with curative potential in CLL (van Gelder et al. 2017; Krämer et al. 2017). AlloHCT can provide long-term disease control even in patients with an unfavorable biological and clinical risk profile, and those having failed pathway inhibitors (Schetelig et al. 2008, 2021; Roeker et al. 2020). The timing of alloHCT should be individually discussed with the patients by taking into consideration the risk of complications after alloHCT and the chances of pharmacological treatment options remaining. Standard risk scores like the HCT-CI or the EBMT-risk score can be used to assess the risk of nonrelapse mortality of an individual patient.

85.4 Remission Induction Prior to Start of the Conditioning Regimen

Large prospective and retrospective studies uniformly show that the results of alloHCT deteriorate if the disease is not controlled at the time of transplant. Thus, alloHCT should be performed in remission of CLL. Apart from experimental approaches, different options exist for remission induction and bridging to alloHCT, basically relying on covalent BTKi, venetoclax, CD20 antibodies, and combinations of these compounds. Non-covalent BTKi may add to the bridging armamentarium as soon as they are labeled (Mato et al. 2023). In addition, the PI3 kinase inhibitor idelalisib can be helpful for bridging albeit its efficacy in double refractory patients is limited (Schetelig et al. 2021)

85.5 Conditioning Regimens

The crucial therapeutic principle of alloHCT in CLL is graft-versus-leukaemia (GVL) activity. Evidence for this comes from the observation that even some patients with refractory disease benefit from alloHCT, a reduced relapse risk in the presence of cGVHD, and the efficacy of immune modulation for the eradication of minimal residual disease (Krämer et al. 2017).

Accordingly, long-term disease control can be achieved with a broad range of conditioning regimens. Current evidence does not allow the definition of one standard conditioning regimen for CLL. The most convincing data supporting alloHCT in CLL come from studies of reduced-intensity conditioning (Krämer et al. 2017; Roeker et al. 2020; Schetelig et al. 2017). The choice of conditioning intensity may vary according to the individual situation. In the presence of comorbidity and chemosensitive disease RIC appear to be more appropriate, whereas high-intensity regimens might be preferable in younger patients with good performance status but poorly controlled disease.

85.6 Outcome After Allogeneic Transplantation for CLL

In a large registry cohort from the pre-pathway inhibitor era, estimated event-free survival, overall survival, and non-relapse mortality

(NRM) 10 years after alloHCT were 28% (95% confidence interval (CI), 25–31), 35% (95% CI, 32–38), and 40% (95% CI, 37–42), respectively (van Gelder et al. 2017). Patients who passed the 5-year landmark event-free ($N = 394$) had a 79% probability (95% CI, 73–85) of surviving the subsequent 5 years without an event. Relapse and NRM contributed equally to late treatment failure. Higher age, lower performance status, unrelated donor type, and unfavorable sex-mismatch have an adverse impact on 2-year NRM. Despite the risks of NRM and relapse/progression the prospect of long-term disease-free survival on average in almost one out of three patients remains an argument to consider allo-HCT especially for younger patients with high-risk CLL. Moreover, NRM appears to be decreased in CLL transplants performed in the pathway inhibitor era (Roeker et al. 2020; Kim et al. 2020; Tournilhac et al. 2021).

85.7 Post-Transplant Minimal Residual Disease Monitoring and Immune Intervention in CLL

In CLL, sensitive MRD quantification (i.e., 1 cell in 10^4 or less) can be obtained by PCR- or flow cytometry-based assays. The decline of the MRD level is often delayed and is closely related to immuno-reconstitution after transplantation. GVL-induced MRD negativity after alloHCT is sustained in the majority of patients and is highly predictive of freedom from relapse. MRD monitoring is a valid instrument for the guidance of pre-emptive immune interventions directed at disease eradication after allo-HCT, such as the tapering of immunosuppression and the use of DLI. The published evidence suggests that CLL is sensitive to timely preemptive immune intervention by modulation of systemic immunosuppression (Krämer et al. 2017).

85.8 Autologous HCT and CAR-T Cell Therapy

Whereas autoHCT does not play a role in current CLL management algorithms except for selected patients with Richter transformation, there is accumulating evidence that CD19-directed CAR-T cells can provide sustained disease control at least in a minority of patients with advanced CLL (Siddiqi et al. 2023). However, as long as there is no labelled CAR-T product available, CAR-T treatment of patients with CLL should be performed only in clinical trials.

85.9 Summary and Perspectives

AlloHCT from related or unrelated donors can induce long-term disease-free survival in patients with high-risk CLL. It is a standard treatment option for patients with high-risk CLL who have failed two pathway inhibitor classes, or who are refractory to one class in the presence of TP53 aberrations and/or a complex karyotype. Generally, alloHCT should be considered before the disease has advanced to a status of complete refractoriness. At the same time, alloHCT should not be recommended for patients who face a higher short-term risk of mortality after transplantation compared to conventional therapy. In the absence of randomized controlled comparisons of these treatment strategies, the outcome of an individual patient has to be predicted based on published data. This requires careful individual assessment of the risk of alloHCT versus continuation of conventional treatment. Patients should be referred to a transplant center once their disease proved refractory to at least one pathway inhibitor in order to get consultation with an expert in the field. Finally, all approved drugs for CLL can also be used for the treatment of post-transplant relapse and further improvements of donor selection, patient care and prevention of complications can be expected, thus, overall outcome after transplantation will continue to improve.

Key Points: AlloHCT in CLL

Indications for alloHCT	• CLL after failure of 2 pathway inhibitor classes • CLL with TP53 aberration and/or complex karyotype after failure of one pathway inhibitor • History of Richter's transformation
Remission induction prior to start of conditioning	Patient who receive alloHCT in remission have a lower risk of relapse. The most promising option for remission induction should be choosen
Donor selection and graft source and GVHD-prophylaxis	No disease-specific criteria have to be considered for the selection of the donor, graft source, or GVHD-prophylaxis
Conditioning	There is no advantage of using a myeloablative regimen over reduced-intensity conditioning
MRD monitoring	MRD-driven immune modulation after alloHCT may improve the likelihood of sustained CLL eradication
Risk factors for non-relapse mortality	• Advanced age • Poor performance status and/or high HCT-CI score • Partially matched as compared to matched donor HCT • Unfavourable donor patients sex constellation
Relapse after alloHCT	Relapse after alloHCT may be treated successfully. To current knowledge, the history of alloHCT does not restrict pharmacological treatment options for patients with relapsed CLL

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